Real-world Data Project

Retail Sales Analysis and Profit Prediction

Domain: Retail | Tools: Python, Pandas, Matplotlib, Seaborn, Scikit-learn

Executive summary

This project analyses retail order data end to end: cleaning, exploration, visualisation and a machine-learning model that predicts whether an order will be profitable. The main finding is that **discounting is the biggest driver of profit**: orders with deep discounts lose money, while low-discount orders earn healthy margins.

- Total sales were $2,665,359 with $453,958 profit, an overall margin of 17.0%.
- December is the strongest month, contributing 14.0% of all sales (seasonal peak).
- Office Supplies has the best margin (23.7%); Furniture has the weakest (5.8%).
- Sub-category 'Copiers' earns the most profit ($145,530); 'Furnishings' earns the least ($3,035).

1. Objective and data

Business question: Which products, regions and pricing decisions drive profit, and can we predict whether an order will be profitable?

The dataset contains 5,060 retail order records (2023-2025) with order date, ship mode, customer segment, region, category, sub-category, sales, quantity, discount and profit.

2. Data cleaning

Step	Method	Result
Duplicates	Dropped exact duplicate rows	60 removed
Missing values	Mode for categories, median for numbers	215 cells filled
Outliers	IQR rule (3x), capped at upper bound	279 values capped
Final dataset		5,000 rows, 0 missing

3. Exploratory analysis

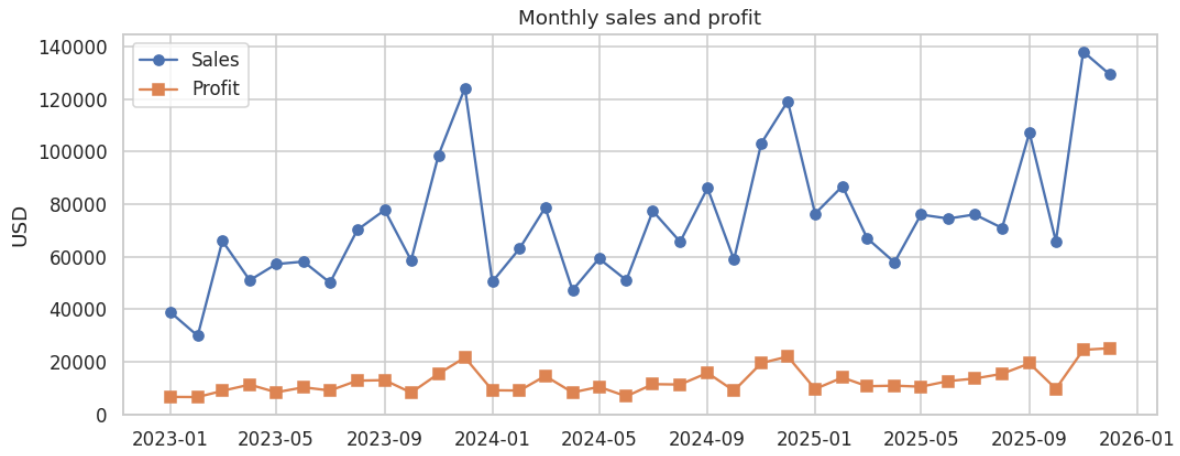


Figure 1: Monthly sales and profit show a clear Q4 seasonal peak.

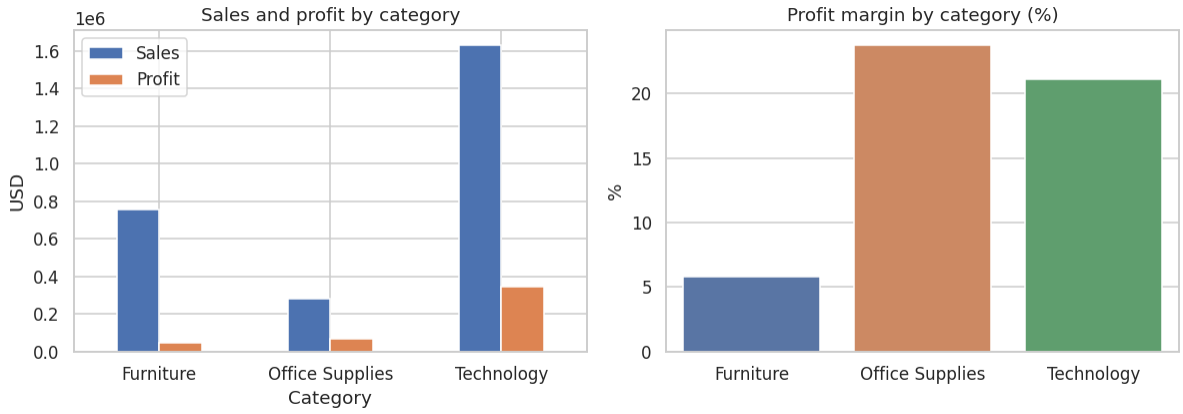


Figure 2: Office Supplies has the strongest margin; Furniture the weakest.

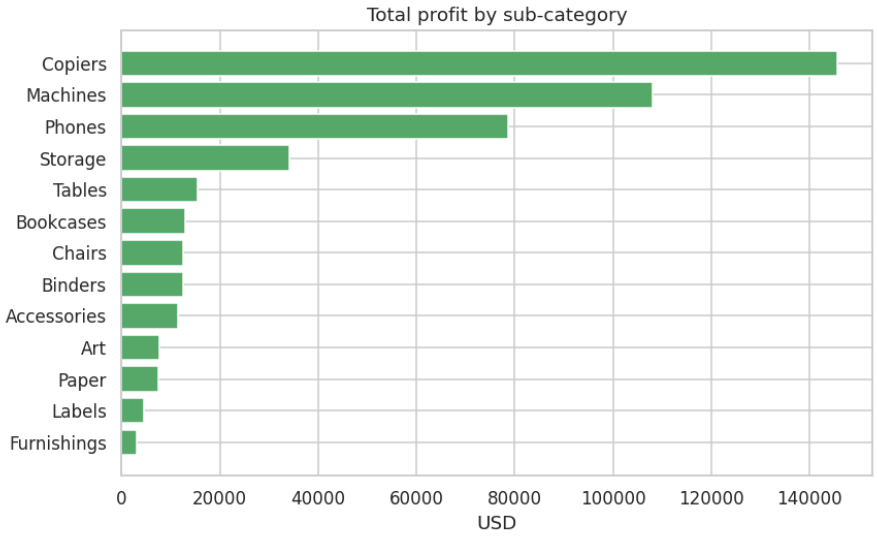


Figure 3: Profit by sub-category.

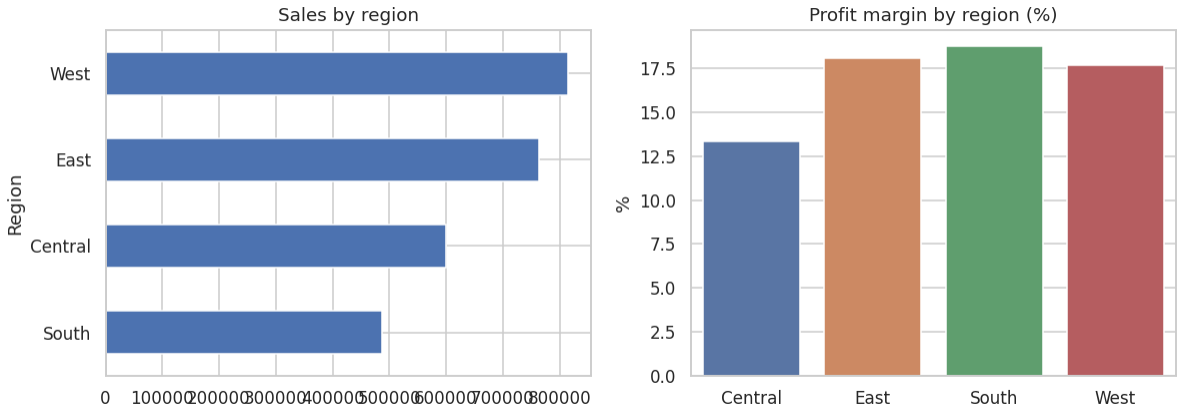


Figure 4: Sales and margin by region.

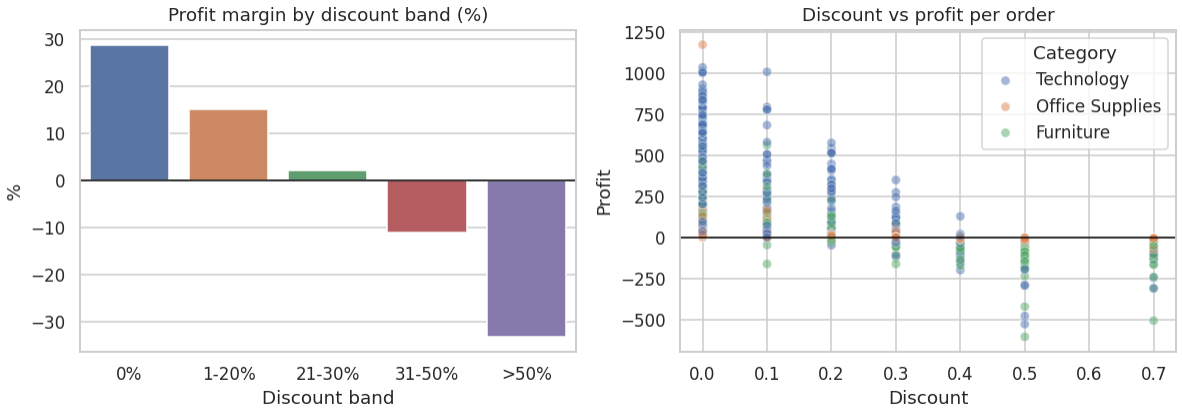


Figure 5: Margins turn negative above about 30% discount.

4. Predictive model

A Logistic Regression baseline and a Random Forest classifier were trained on an 80/20 train-test split to predict whether an order is profitable (profit above zero).

Model	Accuracy	ROC AUC
Logistic Regression	94.8%	0.989
Random Forest	94.9%	0.988
Majority-class baseline	78.5%	-

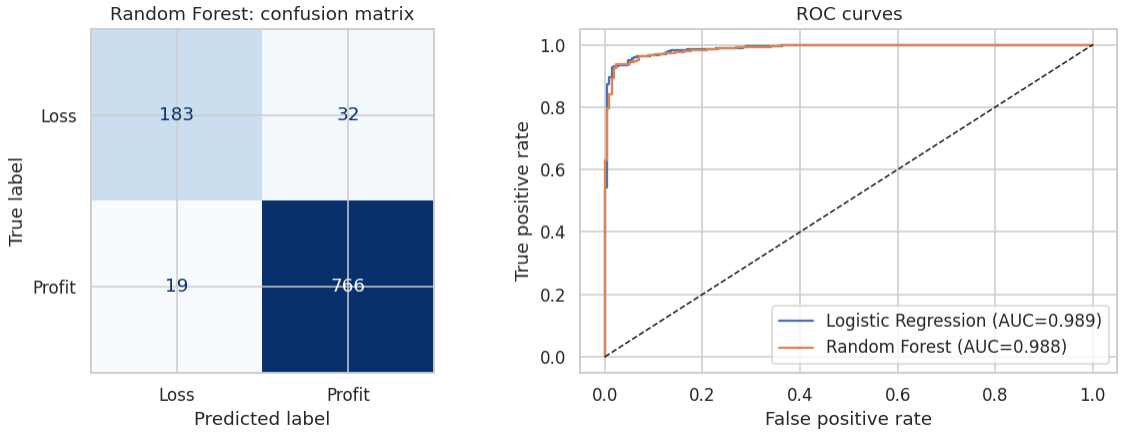


Figure 6: Confusion matrix and ROC curves.

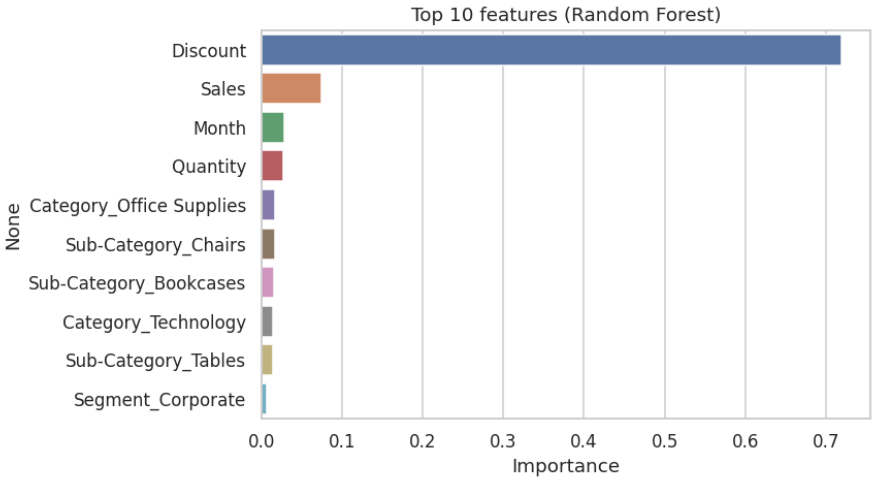


Figure 7: Discount is the most influential feature.

5. Key findings

- Total sales were $2,665,359 with $453,958 profit, an overall margin of 17.0%.
- December is the strongest month, contributing 14.0% of all sales (seasonal peak).
- Office Supplies has the best margin (23.7%); Furniture has the weakest (5.8%).
- Sub-category 'Copiers' earns the most profit ($145,530); 'Furnishings' earns the least ($3,035).
- Region South has the best margin (18.8%); Central the weakest (13.3%).
- Discounts of 40% or more make up 20.2% of orders and produced a net profit of -$49,384 in total (a loss); discount and margin correlate at -0.92.
- Random Forest predicts whether an order is profitable with 94.9% accuracy (baseline 78.5%) and AUC 0.988.
- The most influential features were: Discount, Sales, Month.

6. Recommendations

- Cap discounts at roughly 30% and require approval above that level, since deeper discounts erase the margin.
- Review pricing and discounting in the weakest segments (Furniture, Central region, and 'Furnishings').
- Build inventory and marketing capacity ahead of December and the Q4 season.
- Use the profit-prediction model as a pre-approval check: flag discounted orders likely to lose money before they are confirmed.

7. Limitations

- Findings describe this dataset only; correlation does not prove causation.
- Profit prediction uses order-level attributes; customer history and product costs would improve accuracy.
- **Data note:** the dataset in this package is a simulated Superstore-style retail dataset created for this project. To use a real public dataset, replace *data/retail_sales.csv* with the Kaggle 'Superstore Sales' file (same column names), re-run the notebook, and update the numbers in this report.